Ernest Aboagye

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Employment

Senior Lecturer in Actuarial Science, Purdue University, Indianapolis 2025–present
Department of Mathematics and Statistics

Education

Ph.D., Risk Management and Insurance, Georgia State University 2025
Maurice R. Greenberg School of Risk Science, Robinson College of Business. Minor: Economics.
Dissertation: *Essays on Financial and Actuarial Risk Modeling*. Advisor: Liang Peng

M.S., Mathematics (Actuarial Science), Illinois State University 2017
B.S., Actuarial Science and Statistics, University of Cape Coast, Ghana (Valedictorian) 2014

Professional Designation

Associate of the Society of Actuaries (ASA) 2020
Fellowship (QFI track) modules completed: Enterprise Risk Management, Financial Economics, Social Insurance, Scenario Modeling

Research Interests

Insurance and catastrophe risk: climate and flood risk, public insurance programs (NFIP, federal crop insurance), cyber and data-breach risk, exposure-adjusted loss models

Risk transfer: optimal reinsurance and excess-of-loss contracts, risk measures, estimation risk in contract design

Financial econometrics and systemic risk: contagion and tail dependence, marginal expected shortfall, conditional risk forecasting and backtesting

Statistics of extremes: peaks over threshold, multivariate regular variation, bootstrap inference for tail functionals

Publications

· "A Revisit of the Optimal Excess-of-Loss Contract" (with Vali Asimit, Tsz Chai Fung, Liang Peng and Qiuqi Wang), *European Journal of Operational Research*, 2025.

· "A Contagion Test with Unspecified Heteroscedastic Errors" (with S. I. M. Ko, C. C. Lo, C. Y. L. Hsiao and Liang Peng), *Journal of Economic Dynamics and Control*, 159, 104804, 2024.

Working Papers

· "Is the Past a Reliable Guide to the Flood Tail? Nonstationary Event Losses and the Design of the NFIP's Reinsurance Layer" (sole-authored)

· "Volatility or Contagion? Decomposing Crisis Changes in the Marginal Expected Shortfall of Insurers and Banks" (with Susana Adobea Yamoah)

· "More Often or Bigger? Exposure Growth and Tests for a Shifting Insurance Loss Tail" (with John Koomson and Emmanuel Boamah)

· "Does Index Insurance Fail When It Matters Most? Basis Risk in the Tail of U.S. Area Crop Insurance" (with Agnes Adjoa Anyomi and Siegfried Kafui Anyomi)

· "Claim Counts, Claim Sizes and Excess-of-Loss Pricing under a Common Event Shock" (with Adolph Nii-Armah Okine)

· "When Home Insurers Fail, Do Local Governments Pay? Evidence from Florida" (sole-authored)

· "No Cyber Run: Bank Funding after the MOVEit Mass Data Breach" (with Aaron Kyei Yorke)

· "Testing for a Break in Downside Dependence between Conditional-Mean Errors at a Fixed Quantile" (with Ernest Agyemang Duah and Mohammed Adjieteh)

· "Block-Bootstrap Inference for Changes in Marginal Expected Shortfall with Recorded Returns" (with Ernest Agyemang Duah and Mohammed Adjieteh)

· "Cyber Accumulation Risk under Sparse Incident Data: Conditional Annual Capacity Certification" (sole-authored)

· "Model-Based Conditional Risk Forecasts" (with Liang Peng)

Work in Progress

· "E-Backtesting Expected Shortfall Forecasts under Estimation Risk" (with Qiuqi Wang and Liang Peng)

· "Managing Exposure under Hard Rolling Expected-Shortfall Limits" (sole-authored)

· "Estimation Risk in the Optimal Excess-of-Loss Contract" (with Tsz Chai Fung and Liang Peng)

· "Does Geographic Pooling Diversify Flood Risk? Spatial Tail Dependence and the Tail Index in U.S. Flood Insurance" (with Shimeng Huang)

· "Bounded Tails, Correlated States: Who Holds Federal Crop Insurance Catastrophe Risk under the Standard Reinsurance Agreement?" (sole-authored)

· "More Often, Not Bigger: The Shifting Tail of U.S. Health Data Breaches and the Third-Party Channel" (with Adolph Nii-Armah Okine)

Other Publications

· "Optimizing Travel Insurance Purchase Detection Using Predictive Models" (with B. Borketey and K. Danquah), *International Journal of Automation, Artificial Intelligence and Machine Learning*, 2024.

· "Predictive Accuracy of the Probability of Credit Card Default Using Machine Learning" (with B. Borketey, D. Borketey and K. Danquah), *International Research Journal of Modernization in Engineering Technology and Science*, 2022.

Presentations and Conferences

Purdue University, Departments of Mathematics and Statistics — Feb 2025
A Revisit of the Optimal Excess-of-Loss Contract

Bentley University, Department of Mathematical Sciences — Dec 2024
A Revisit of the Optimal Excess-of-Loss Contract

Illinois State University, Actuarial, Insurance and Statistics Seminar — Oct 2024
A Revisit of the Optimal Excess-of-Loss Contract

American Risk and Insurance Association (ARIA) Annual Meeting, Denver (participant) — 2024

27th International Congress on Insurance: Mathematics and Economics (participant) — 2024

ARIA Annual Meeting, virtual (participant) — 2021

International Association of Black Actuaries Annual Conference (participant) — 2016–2024

Teaching

Instructor of record. Ratings are mean instructor evaluations.

Purdue University

MA 37300 Financial Mathematics	5.0/5.0	S2026
MA/STAT 41600 Probability	4.9/5.0	S2026
CS/STAT 24200 Introduction to Data Science	4.4/5.0	S2026
MA 15800 Precalculus	4.6/5.0	F2025

Georgia State University

AS 3230 Financial Mathematics	4.2/5.0	F2023
AS 4140L Probability Lab	4.9/5.0	F2022, S2023, F2024, S2025
AS 3230L Financial Mathematics Lab	4.9/5.0	F2022, F2024, S2025

Illinois State University

MAT 113 Mathematical Reasoning Lab (instructor)	2015–2016
Teaching assistant: MAT 280, MAT 480, MGT 100, MGT 227	2015–2017

University of Cape Coast

Teaching assistant: algebra, calculus, interest theory, life contingencies	2014–2015

Fellowships, Grants and Awards

Spencer Educational Foundation Doctoral Fellowship ($10,000)	2024–2025
Center for the Economic Analysis of Risk (CEAR) Fellowship ($15,000 total)	2022–2025
S. S. Huebner Foundation Fellowship ($16,000 total)	2021–2025
SOA James C. Hickman Scholar ($40,000 total)	2021–2023
GSU Graduate Fellowship and Assistantship	2020–2025
GSU Greenberg School of Risk Science Research and Travel Grant ($3,637)	2024
International Association of Black Actuaries Scholar ($17,000 total)	2016, 2021–2024
Best Paper Award, ISU Annual Statistical Project Competition ($2,500)	2017
Valedictorian, University of Cape Coast	2014
Prudential Life Insurance Ghana Best Actuarial Student Award	2014

Professional Service

Referee: *Statistical Papers*	2026
Judge, Stevie Awards, Women in Business category	2024–2025
Technology Committee, International Association of Black Actuaries	2018–2020

Industry Experience

Actuarial Analyst, Retirement Consulting, Willis Towers Watson	2019–2020
Actuarial Analyst, Life Actuarial Consulting, Milliman	2018
Actuarial Consultant, Life Pricing Analytics, New York Life Insurance Company	2017
Actuarial Intern, Commercial Lines Pricing, American Family Insurance	2017

Skills

R, Python, MATLAB, Stata, SAS, Excel VBA, LaTeX; actuarial platforms Prophet, MG-ALFA, Quantify. Languages: English (fluent), Twi, Ga, Fante.

References

Available on request.